

Probability

Random Experiment

The experiment for which the set of all different possible results or outcomes are known a priori and which is such that it is impossible to predict which one of this set will occur at any particular performance of the experiment is called random experiment.

Sample Space

Event:

Elementary Event, Composite Event, Certain Event, equally likely event, Mutually Exclusive Event, Dependent Event, Independent Event.

Trial

Exhaustive

Classical Definition of Probability:

Let E be a random experiment such that its event space S contains finite number of event points say n all of which are known to be equally likely. If any event A connected with E has m(A) favorable cases the probability of A is a pure number and is denoted as P(A) then

$$P(A) = \frac{m(A)}{n}$$

Statistical or Empirical Definition of Probability:

If a trial is repeated a number of times under essentially homogeneous and identical conditions then the limiting value of the ratio of the number of times the event happens to the number trials, as the number of trials becomes indefinitely large is called the probability of happening of the event, i.e.,

$$P(A) = \lim_{n \rightarrow \infty} \frac{m(A)}{n}$$

Axiomatic Definition or Kolmogorov's Definition of Probability:

Let E be a random experiment described by the sample space S and A be any event points belongs to S, i.e., $A \in S$. Then probability of A is a pure number denoted by P(A) such that the following axioms are satisfied:

i). $0 \leq P(A) \leq 1$,

ii). The probability of certain event is one, i.e., $P(S)=1$

iii). If $A_1, A_2, A_3, \dots$ be finite or infinite sequence of pairwise mutually exclusive events, i.e., $A_i \cap A_j = \emptyset$ where $i \neq j, i, j = 1, 2, 3, \dots$ then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_n \cup \dots) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_n) + \dots$$

Theorems:

1. $\text{Prob}(A) = 1 - \text{Prob}(\bar{A})$
2. $\text{Prob}(\emptyset) = 0$

Addition Law:

If A and B be any event then

$$\text{Prob}(A \cup B) = \text{Prob}(A) + \text{Prob}(B) - \text{Prob}(A \cap B)$$

Generalized Addition law

$$\text{Prob}\left(\bigcup_{i=1}^n A_i\right) = \sum_{i=1}^n \text{Prob}(A_i) - \sum_{1 \leq i < j \leq n} \text{Prob}(A_i \cap A_j) + \sum_{1 \leq i < j < k \leq n} \text{Prob}(A_i \cap A_j \cap A_k) - \dots + (-1)^{n-1} \text{Prob}(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n)$$

Boole's Inequality:

$$\text{Prob}(A \cup B) \leq \text{Prob}(A) + \text{Prob}(B)$$

Conditional Probability

Let A & B are two events associated with an experiment E. The conditional probability of event B, given that A has occurs is denoted as

$$\text{Prob}(B/A) = \frac{\text{Prob}(A \cap B)}{\text{Prob}(A)}, \text{ provided } \text{Prob}(A) \neq 0 \text{ and}$$

$$\text{Prob}(A/B) = \frac{\text{Prob}(A \cap B)}{\text{Prob}(B)}, \text{ provided } \text{Prob}(B) \neq 0$$

$$\text{Therefore, } \text{Prob}(A \cap B) = \text{Prob}(A/B) \text{Prob}(B) = \text{Prob}(B/A) \text{Prob}(A)$$

Independence: The two events A & B are said to be independent if the occurrence of A is not influenced by the occurrence of B and vice versa., i.e. $\text{Prob}(A \cap B) = \text{Prob}(A) \text{Prob}(B)$

Multiplicative Law

$$\text{Prob}(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_n) = \text{Prob}(A_1) \text{Prob}(A_2 / A_1) \text{Prob}(A_3 / A_1 \cap A_2) \times \dots \times \text{Prob}(A_n / A_1 \cap A_2 \cap A_3 \cap \dots \cap A_{n-1})$$